Assignment 4

Exercise 1: Ring Communication

Processes (p)	Time (Tp)	Speedup (Sp)	Efficiency (Ep)
1	0.000070	1.00	100%
2	0.000048	1.45	72%
4	0.000085	0.82	20%
6	0.017500	0.004	0.07%

The Ring Communication program shows reduced performance as the number of processes increases because communication overhead becomes more significant. As more processes participate in the ring, each process spends additional time sending and receiving messages, which increases overall execution time. This overhead gradually dominates the computation, leading to lower speedup and efficiency at higher process counts.

Exercise 2: Parallel Array Sum

Processes (p)	Time (Tp)	Speedup (Sp)	Efficiency (Ep)
1	0.000007	1.00	100%
2	0.000025	0.28	14%
4	0.000042	0.16	5%

Exercise 3: Max-Min

Processes (p)	Time	Speedup	Efficiency
1	0.000003	1.00	100%
2	0.000025	0.10	5%
4	0.000120	0.02	0.5%
6	0.030000	0.00007	0.002%

Execution time increases due to synchronization overhead among processes. In parallel programs, processes often need to coordinate and wait for each other at certain points, which introduces delays. This waiting time adds extra overhead and reduces the overall performance of the system.

Exercise 4: Parallel Dot Product

Processes (p)	Time (Tp)	Speedup (Sp)	Efficiency (Ep)
1	0.000007	1.00	100%
2	0.000017	0.41	21%
4	0.024000	0.00025	0.007%

Small data size leads to poor scalability because communication overhead becomes dominant compared to computation. Since the amount of work is limited, the time spent in communication between processes is relatively high. This reduces the benefits of parallel execution and results in low performance improvement.